

David Robert

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Portfolio:

www.davidrobert.computer

Creative Technologist & Interactive Systems Designer

I apply emerging technologies to interactive experiences in the physical world.

Experienced across art studios, design agencies, and research labs.

My work has been shown internationally and engaged by thousands of participants in public space.

Focused on real-time interactive systems, generative & AI pipelines, and physical computing.

EXPERIENCE

- Creative Technologist (R&D)

Antimodular Research Inc.,

Atelier Lozano-Hemmer

Montreal QC

August 2023 - present
- Built and deployed real-time interactive systems for public artworks by Rafael Lozano-Hemmer, turning conceptual proposals into durable interfaces that have been used by tens of thousands of participants, built using TouchDesigner, Python, Arduino, and more.
 - Led development of multimodal AI systems spanning computer vision, audience tracking, semantic retrieval, generative text, audio analysis, and spatialized sound for large-scale public installations. For Undercurrents, this included a real-time system that listens to visitors and responds with contextually selected audio.
 - Integrated and networked installation hardware including RealSense depth cameras, LiDAR, IP and USB cameras with computer vision, ArtNet and DMX lighting at scale, projectors, and Dante audio. Linked multi-computer setups over OSC, MQTT, HTTP, and NDI, and chose media formats and codecs for reliable playback.
 - Installed and supervised exhibitions internationally, including month-long deployments in Mexico City and Abu Dhabi. Worked directly with curators, collectors, and museum teams, and with venues and outside contractors on power, safety, and fabrication.
 - Contributed to project scoping, hardware procurement, timeline estimation, and technical planning, with autonomy over language choice and project architecture. Kept work in Git; wrote Windows startup, shutdown, and batch automations; and documented every artwork with guides and manuals for technical teams and clients.

- Media Artist in Residence

Fabrica Research Centre

Treviso TV, Italy

September 2024 - December 2024
- Selected for an international artist residency in which I developed and exhibited an interactive installation exploring AI mimicry and vocal embodiment, built with TouchDesigner, Python, and cloud-based APIs.
 - Developed the work through critique and cross-disciplinary exchange with an international cohort, centered on the theme of Ruralism.

- Creative Technologist / Lab Assistant

Design + Technology LAB,

Toronto Metropolitan University

Toronto ON

September 2021 - April 2022

September 2022 - April 2023
- Assisted in day-to-day operations of the lab, including maintenance and operation of lab equipment. This includes 3D printing, laser cutting, XR, and robotics.
 - Created an interactive 3D projection-mapping installation for DesignTO 2022 using TouchDesigner, a KUKA robot arm, and full-stack web development.

EDUCATION

BA in Media Production

Concentration in Digital Media

4.27 / 4.33 GPA

Toronto Metropolitan University

Toronto ON

September 2019 – April 2023

PUBLICATIONS

David Bouchard, Cintia Cristia, David Robert, Michael Bergmann. 2022. Augmented Symphony: An augmented reality application for immersive music listening. In *Proceedings of EVA London 2022*. Computer Arts Society, London, England, UK.

EXHIBITIONS

David Robert. I SURRENDERED MY BODY AND I SUCCUMBED TO THE BEAST. At *Until the Very End*. Treviso, Italy. *Contribution: Artist*.

David Robert, Alex Verni, Design + Technology LAB. 2022. Assembly Line. At *DesignTO 2022*. Toronto, Canada. *Contribution: Lead Developer and Designer*.

David Robert, Alex Verni, Stacy Cernova, Anthony Baloukas. 2022. Moth Melody. At *Ontario Science Centre*. Toronto, Canada. *Contribution: Lead Developer*.

SKILLS

Programming

- | | |
|----------|-------------|
| -Python | -JavaScript |
| -Node.js | -React |
| -GLSL | |

Interactive Media

- | | |
|----------------|----------------|
| -TouchDesigner | -Unreal Engine |
| -Unity | -Three.js |
| -ComfyUI | |

Systems & Networking

- | | |
|------------|-------------|
| -OSC | -MQTT |
| -NDI | -WebSockets |
| -REST APIs | -Git |

Physical Computing

- | | |
|---------------|-----------------|
| -Arduino | -Raspberry Pi |
| -DMX / ArtNet | -Depth cameras |
| -LiDAR | -Circuit design |
| -3D printing | -Laser cutting |

Research Assistant [AI] Technology Research in Performance Lab, Toronto Metropolitan University Toronto ON September 2022 - April 2023	▫ Researched human–AI co-performance, building experimental systems that merged live theatre with generative AI. ▫ Built a real-time theatrical AI-improvisation system using cloud-based AI tools, serving as an early prototype of interactive dialogue with AI in public performance.
Teaching Assistant [Physical Computing] Toronto Metropolitan University Toronto ON September 2022 - December 2022	▫ Assessed students and gave feedback on projects for RTA 321: Intro to Tangible Media. ▫ Supported students with Arduino programming and simple circuit design.
Creative Developer Intern Jam3 Toronto ON May 2022 - August 2022	▫ Built a multi-user online 3D space in Three.js with persistent data (React, Node, MongoDB, WebSockets), showing how small businesses could interact with the Metaverse. Delivered as a client-facing prototype. ▫ Worked in a fast-paced, autonomous intern team of developers, designers, and a coordinator.
Research Assistant [AR] Toronto Metropolitan University Toronto ON July 2021 - April 2022	▫ Developed an AR app using Unity [C#] centred on enhancing digital delivery of orchestral music through spatial and interactive audio. ▫ Co-authored peer-reviewed paper accepted to EVA London 2022.
ADDITIONAL EXPERIENCE	
Undergraduate Researcher [Sim Developer & HRI Designer] Synaesthetic Media Lab, Toronto Metropolitan University Toronto ON September 2022 - December 2022	▫ Collaborated with a joint university research team to prototype interactive, networked “boat-drones” for live performance environments. ▫ Designed and programmed a simulation environment in Unreal Engine 5 (C++) and built real-time communication systems (Python server, WebSockets, I2C) to synchronize physical and virtual performance elements.
Interactive Developer Ontario Science Centre Toronto ON January 2022 - April 2022	▫ Designed and delivered a projection-based audio installation that engaged families and children, turning abstract tech into playful, intuitive interaction. Ensured reliability for constant public use.
Undergraduate Researcher [Unreal AR Developer] Synaesthetic Media Lab, Toronto Metropolitan University Toronto ON January 2021 - April 2021	▫ Designed and developed an AR HoloLens 2 application in Unreal Engine 4 [Blueprints]. ▫ The app supports training for medical workers through the manipulation of medical models with gestures and tools.
Fabrication Technician Philip Beesley Studio, Living Architecture Systems Group Toronto ON February 2020 - March 2020	▫ Fabricated components for an interactive installation that was part of the 2021 Venice Biennale of Architecture.